

# FYKE SIMON V. TONEL

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## Education

**STI College Caloocan** 2022 – 2026

Bachelor of Science in Computer Science Caloocan City, Metro Manila

- Cumulative GWA: 1.25 (Latin Honor: Magna Cum Laude)
- Best Thesis, Bachelor of Science in Computer Science (Campus-wide – ICTE Symposium), 2024-25
- Best Methods of Research Project, Bachelor of Science in Computer Science (Campus-wide – ICTE Symposium), 2024–25
- Entrant, Codefest (Local – Tagisan ng Talino), 2024–25
- President’s Honor (STI Academic Center – Academic Honors), 2022 – Present
- Alumni Scholar (Awarded for High Honors in Senior High; 50% scholarship in college)

**STI College Caloocan** 2020 – 2022

Senior High School (Information & Communications Technology) Caloocan City, Metro Manila

- Graduated with High Honors
- Completion of Computer Programming – Java and .NET Technology (IT in Mobile App & Web Development)

## Experience

**Back-End Developer (Intern)** January 2026 – February 2026

Inspire Holdings Inc.

- Built and deployed a NestJS payment backend on AWS (Lambda, API Gateway, RDS), integrating secure UnionBank APIs.
- Reduced **p95** latency from **420ms to 180ms (–57%)** and CPU usage by **35%** using Redis caching and token-bucket rate limiting.
- Enforced **API security via HMAC signing** and request validation.
- Delivered **CI/CD pipelines** and **40+ unit tests** using GitHub Actions and Jest.

**Software Developer (Intern)** February 2026 – March 2026

Socia I.T. Solutions

## Projects

**LootBX Mobile: Real-Time Streaming & Interactive Rewards Platform** February 2026

React Native, Expo, TypeScript, JavaScript, MongoDB

Full-Stack Developer

- Developed API integrations supporting **real-time data flow** for a mobile streaming and rewards platform.
- Implemented state management logic during migration from **web-based architecture to unified mobile application**.
- Contributed to full-stack development of a **low-latency streaming system**, improving responsiveness and scalability.
- Applied TypeScript and modern backend patterns to improve **code reliability and maintainability**.

**Papa Burger Ecosystem: Multi-Platform Restaurant Operations System** February 2026 – March 2026

React TSX, PHP, PostgreSQL, Next.js, Flutter, Dart

Full-Stack Developer

- Built front-end architecture for a **multi-module restaurant system** integrating POS, driver, and admin platforms.
- Developed core modules including **POS system, driver portal, and franchising interface**.
- Designed reusable UI components and state management for **scalable and maintainable frontend systems**.
- Integrated backend APIs to support **order processing, transactions, and operational workflows**.

## Projects

**Flick: Music Player with Custom UAC 2.0** January 2026 – Present

Flutter, Dart, Rust (DSP, Audio Processing), and Kotlin

Lead Developer – Maintainer

- Engineered **four Rust-based audio engines** enabling direct USB output and bypassing Android audio pipeline constraints.

- Architected a hybrid differential music library scanner using **Android MediaStore**, **Rust parallel traversal**, and **fingerprint caching**, reducing scan times from **~11s to 85–328ms (up to 34× faster)** across **1,000+ track SD-card libraries**.
- Optimized music library processing (scanning, metadata extraction, indexing), improving **performance on large libraries**.
- Open-source project with **2000+ installs/downloads** across **GitHub and Play Store distributions**.

#### **br41ndmg: High-Fidelity Audio Resampling Library**

January 2026 – Present

Rust (DSP, Audio Processing)

Lead Developer – Maintainer

- Implemented **polyphase sinc-based resampling** for high-accuracy sample rate conversion.
- Developed configurable **FIR filters** (Hann, Hamming, Blackman, Kaiser) for precise frequency response control.
- Achieved **>100 dB stopband attenuation** and **<0.1 dB passband ripple**.
- Validated DSP performance using impulse, sine, and sweep testing with Criterion benchmarks
- Open-source project with **70+ installs/downloads on crates.io library distributions**.

#### **PASADA: Ride-Hailing & Fleet Management Ecosystem**

December 2024 – November 2025

Flutter, TypeScript (Bun/Deno), React, Supabase, and Google Cloud Platforms

Lead Full-Stack Developer

- Architected a multi-platform system connecting **4 applications** (Passenger, Driver, Admin, Web) into a unified transport ecosystem.
- Developed an analytics pipeline generating **7-day predictive forecasts** for booking demand and route traffic.
- Integrated Gemini API to transform **real-time and predictive data into actionable summaries** for users and fleet managers.
- Developed a Next.js web platform with a **live ride-tracking system** using MapLibre GL and OpenStreetMap.
- **Validated platform scale and UX** through live field testing with **132 active passengers** and **32 assigned modernized jeepney drivers**.

#### **Mochi: Family-Shared AI Companion with Persistent Memory System**

April 2026

Flutter, Node.js, SQLite, llama.cpp, Gemini API, Termux, and Kotlin

Lead Developer

- Designed a multi-user AI system with **persistent memory, mood tracking, and relationship state modeling**.
- Built a hybrid inference pipeline combining **local LLM (llama.cpp)** and **cloud fallback (Gemini API)**.
- Developed backend services for **conversation history, memory injection, and state management**.
- Deployed on a low-cost infrastructure (Termux + Cloudflare Tunnel) for **continuous availability**.

#### **Latch: Advanced Media Hiding & Security Application**

November 2025

Flutter, Dart, and Kotlin

Lead Developer – Maintainer

- Developed an Android media vault with **biometric authentication, PIN protection, and AES-256 GCM/CTR dual-engine encryption with PBKDF2 and Argon2id**.
- Implemented security features including **decoy vault and forced-access protection mechanisms, local back-up and password vaults**.
- Built a full media pipeline (import, compression, secure storage, recovery) supporting **multiple file types**.
- Open-source project with **200+ installs/downloads** across **GitHub and Play Store distributions**.

### **Skills**

#### **Core Languages**

Dart, Rust, Kotlin, TypeScript, Java, SQL

#### **Systems & Backend Engineering**

Node.js (NestJS, ExpressJS), Bun, Deno

PostgreSQL, SQLite, Supabase, Redis, QuestDB, Upstash

AWS, GCP, Docker, Cloudflare

WebSockets, API design, RBAC, rate limiting

#### **Mobile & Frontend Development**

Flutter, React, React Native, Vite, Next.js, Remix

HTML, Tailwind CSS, MapLibre GL JS

#### **Low-level & Performance**

Rust (systems programming, DSP, audio processing)

FFI (Flutter ↔ Rust), real-time pipelines, memory optimization

**Cloud & Infrastructure**

Docker, Vercel, Fly.io, Cloudflare, GCP  
Event-driven systems (SQS/SNS), caching, deployment pipelines

**AI & Applied ML**

llama.cpp, Gemini API, LM Studio

**Tools & Environments**

Linux (Arch Linux and CachyOS), Neovim, Git, GitHub, GitLab, Postman  
Android Studio, IntelliJ IDEA, RustRover, Zed  
Figma

**Certifications**

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|------------------------------------------|-----------|
| <b>Learn Dart: First Step to Flutter</b> | July 2024 |
| Educative.io                             |           |

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| <b>Java Fundamentals</b> | June 2023 |
| Oracle                   |           |